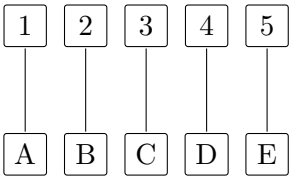
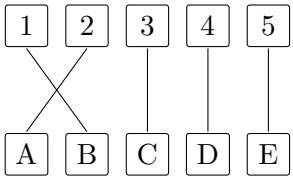
Queues run from left to right. A complete match gives each customer exactly one partner in the other queue, with nobody left out. It keeps order when any two customers and their partners appear in the same order.

Labels only identify customers. Partners may have different labels; identical labels in two queues do not have to be paired.

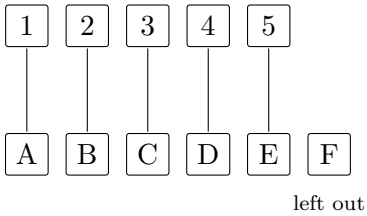
Complete match keeps order



Complete match does not keep order



5 customers and 6 no complete match possible



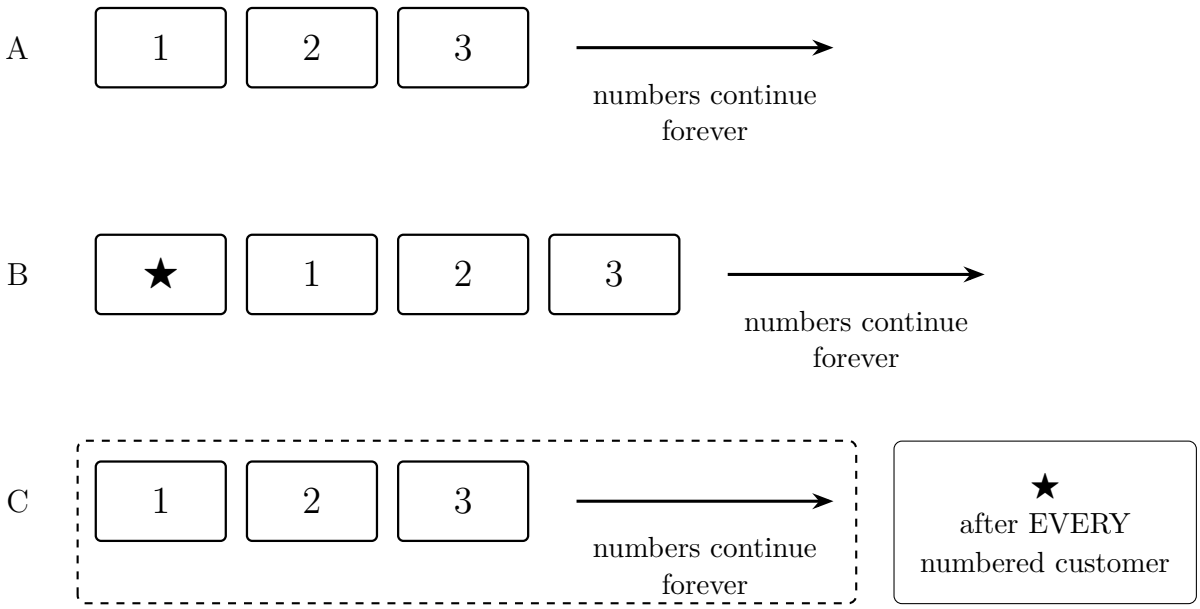
Problem 1: Find every match between these two queues. Which matches keep order? Use cards and yarn to test your matches.

R1 R2 R3

B1 B2 B3

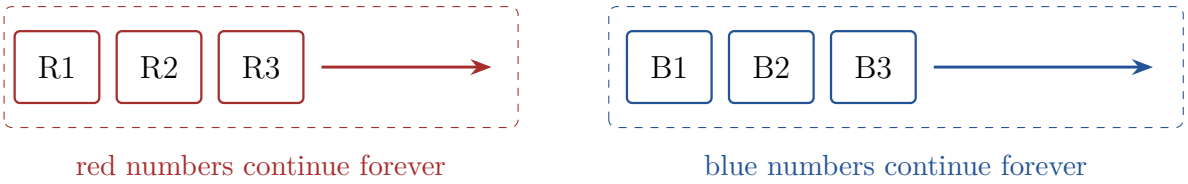
An arrow marked “numbers continue forever” represents all later numbered customers.

Problem 2: Compare A with B, then A with C. Can you match every customer? Can a match keep order? Give a rule for every customer, or explain why it is impossible.



Problem 3: Compare D with E. Can you match every customer? Can a match keep order? Give a rule for every customer, or explain why it is impossible.

D

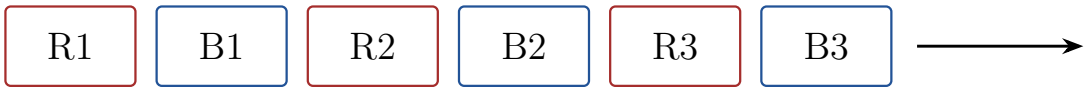


red numbers continue forever

blue numbers continue forever

The whole blue row starts after EVERY red customer.
There is no last red customer.

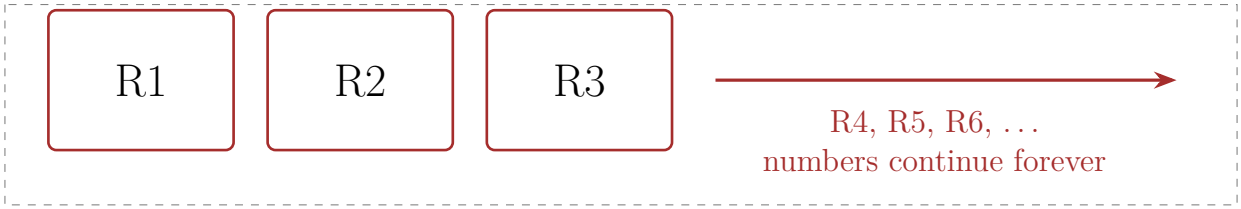
E

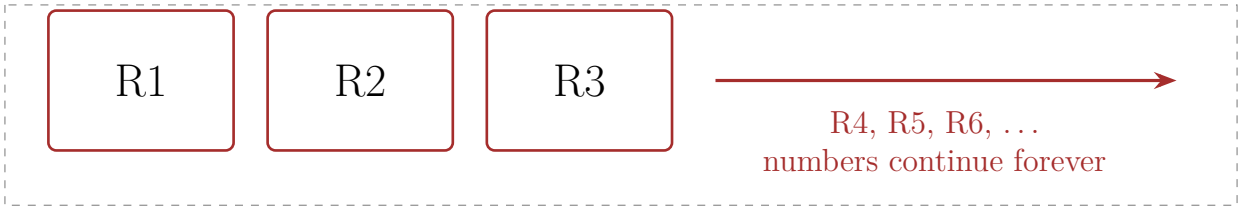


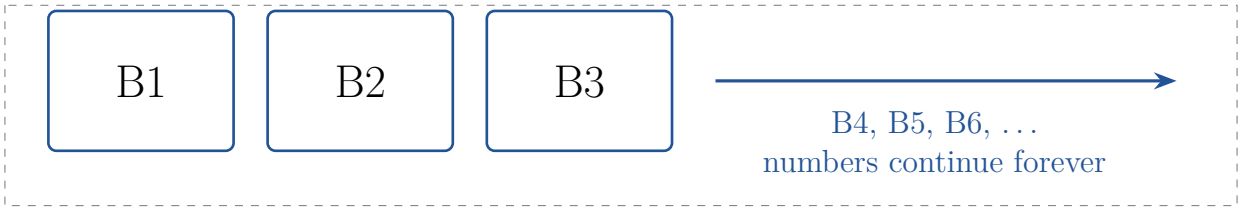
R4, B4, R5, B5, . . .
pairs continue forever

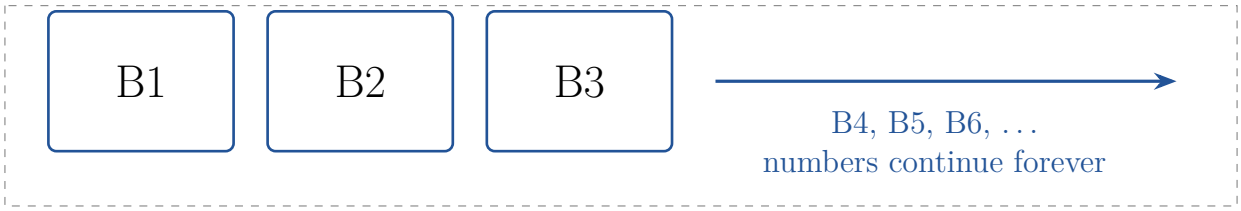
Problem 4: Arrange one or two endless numbered rows and up to two ★ customers on the table. A star or new row following a whole endless row comes after EVERY customer in that row.

Find two different arrangements that can be matched while keeping order. Find two that can be matched only if order may change. Sketch your arrangements in the boxes. Give matching rules or explain the obstruction.

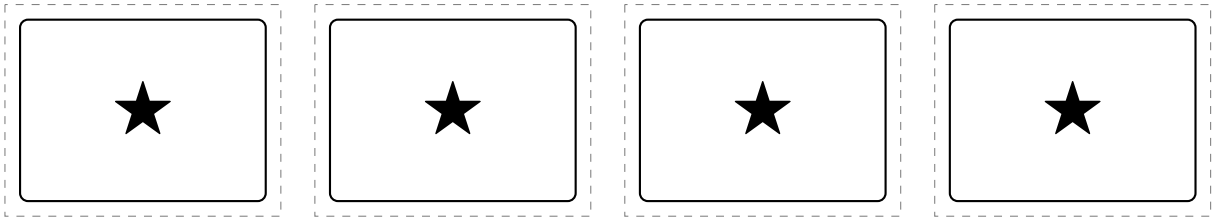

R1
R2
R3
R4, R5, R6, . . .
numbers continue forever


R1
R2
R3
R4, R5, R6, . . .
numbers continue forever


B1
B2
B3
B4, B5, B6, . . .
numbers continue forever


B1
B2
B3
B4, B5, B6, . . .
numbers continue forever

R1	R2	R3	R4
R5	R6	R7	R8
R9	R10	R11	R12
B1	B2	B3	B4
B5	B6	B7	B8
B9	B10	B11	B12



This customer comes after EVERY customer on the earlier strip.

This customer comes after EVERY customer on the earlier strip.

This whole row starts after EVERY customer on the earlier strip.

This whole row starts after EVERY customer on the earlier strip.